

From Monocular Visual Odometry to Stereo Visual Odometry (Metric Scale) on TUM VI

Project Guide for Master's Course in Computer Vision and Robotics

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Abstract—This project guide specifies the course project on implementing classical, geometry-based monocular visual odometry (VO) and extending it to metric stereo VO using the TUM VI benchmark dataset. Students will develop a complete pipeline in Python, beginning with feature-based monocular VO on the left camera and progressing to metric stereo VO leveraging calibrated baseline and disparity.

I. INTRODUCTION AND MOTIVATION

Visual odometry (VO) estimates camera ego-motion from image sequences, forming the backbone of autonomous navigation systems in robotics, autonomous driving, and augmented reality [1]. Monocular VO suffers from inherent scale ambiguity, requiring arbitrary initialization and suffering from drift accumulation over long trajectories. Without metric scale, monocular VO cannot directly support obstacle avoidance or path planning in physical space. VO is also commonly positioned as a building block within broader vision-based SLAM systems [2].

Stereo visual odometry (SVO) resolves these limitations by exploiting a calibrated stereo rig with known baseline B . Synchronized left–right images enable direct metric depth estimation via disparity, yielding metric-scale trajectories and improved robustness to scale drift [3], [4]. This project focuses on classical, geometry-based estimation, avoiding deep learning to build fundamental understanding of epipolar geometry, triangulation, and bundle adjustment.

The TUM VI dataset provides a rigorous benchmark for evaluating these systems [5].

II. DATASET AND EXPERIMENTAL SETUP

The TUM VI benchmark comprises a hand-held rig with global-shutter stereo cameras (512×512, 20 Hz), a 200 Hz IMU, and high-precision motion capture ground truth [5]. For this project, students utilize stereo images and calibration; the IMU is available for optional investigation but not required for the baseline stereo VO.

Mandatory sequences are:

- Room2: full ground truth available.
- Corridor3: start–end ground truth only.
- Outdoors5: start–end ground truth only.

Data is accessible via ROS bags or TUM-format exports. Students must use the provided Kalibr calibration (intrinsics, stereo extrinsics T_{lr} , camera–IMU extrinsics, and photometric calibration); ad-hoc recalibration is forbidden. Report hardware (CPU, RAM), OS, Python version, OpenCV version,

random seeds (e.g., `np.random.seed(42)`), and runtimes to ensure reproducibility.

III. PROJECT TASKS AND GROUP ORGANIZATION

Students may work individually or in pairs (maximum 2). Pairs must divide tasks equitably (e.g., one student on monocular VO frontend, the other on stereo backend and evaluation), document contributions in the report, and demonstrate shared understanding during defense.

The project consists of two stages:

- 1) **Baseline Monocular VO:** Implement feature-based VO using only the left camera, producing up-to-scale trajectories.
- 2) **Metric Stereo VO:** Extend the pipeline to use calibrated stereo pairs, producing metric-scale trajectories via disparity-based depth.

Deep learning is strictly prohibited. Forbidden methods include CNN-based features (SuperPoint, SuperGlue), learned descriptors, neural depth estimation, or pose regression networks. Allowed tools: Python 3.x, OpenCV, NumPy, SciPy, Matplotlib, Open3D. Optional ROS for bag playback. Bonus marks are awarded for partial or complete C++ implementations with documented build systems and reproducible scripts.

IV. BASELINE MONOCULAR VISUAL ODOMETRY

Students must select one classical VO paper or tutorial (e.g., [1]) as their conceptual base, justify the choice, and map each pipeline component to that reference.

A. Feature Detection and Matching

Detect ORB or SIFT features in each frame. Match consecutive frames using descriptor distance with a ratio test and optional cross-check. Verify matches with RANSAC for essential matrix estimation.

B. Epipolar Geometry and Initialization

The essential matrix $\mathbf{E}$ encodes the relative pose between two calibrated views. The epipolar constraint relates corresponding normalized points $\mathbf{x}, \mathbf{x}'$:

$$\mathbf{x}'^\top \mathbf{E} \mathbf{x} = 0, \quad (1)$$

where $\mathbf{x} = [x, y, 1]^\top$ in normalized image coordinates [3].

The essential matrix decomposes as:

$$\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}, \quad (2)$$

where $\mathbf{R} \in SO(3)$ is rotation, $\mathbf{t} \in \mathbb{R}^3$ is translation (up to scale), and $[\cdot]_{\times}$ denotes the skew-symmetric matrix [3].

Decompose $\mathbf{E}$ into four pose hypotheses; disambiguate via cheirality (positive depth) checking. Triangulate an initial map; set initial scale arbitrarily (e.g., $\|\mathbf{t}\| = 1$).

C. Tracking and Optimization

Track features across frames using optical flow or PnP with RANSAC. Refine pose via motion-only bundle adjustment minimizing reprojection error:

$$\mathbf{x} = \pi(\mathbf{T}\mathbf{X}), \quad (3)$$

where $\mathbf{T} \in SE(3)$ is the camera pose, $\mathbf{X} \in \mathbb{R}^3$ is a 3D point, and π is the projection function.

Export up-to-scale trajectories in TUM format (timestamp t_x t_y t_z q_x q_y q_z q_w).

V. STEREO VISUAL ODOMETRY: MODELING AND PIPELINE

Students must select a real stereo VO or stereo SLAM reference (e.g., [4], [6], [7]) as their conceptual base.

A. Stereo Geometry

A calibrated stereo rig provides two rectified images with known baseline B . For a point with disparity d (in pixels), the metric depth Z in the left camera frame is:

$$Z = \frac{fB}{d}, \quad (4)$$

where f is the focal length in pixels [3].

Given image coordinates (u, v) and disparity d , the 3D point $\mathbf{X} = [X, Y, Z]^T$ in the left camera frame is:

$$\mathbf{X} = \frac{Z}{f} \begin{bmatrix} u - c_x \\ v - c_y \\ f \end{bmatrix} = \frac{B}{d} \begin{bmatrix} u - c_x \\ v - c_y \\ f \end{bmatrix}, \quad (5)$$

where (c_x, c_y) is the principal point [3].

B. Stereo VO Pipeline

Per-frame processing:

- Rectify stereo pair using calibration (if not pre-rectified).
- Compute disparity map via block matching (OpenCV StereoBM) or semi-global matching (StereoSGBM).
- Construct 3D point clouds in left camera frame using (5).

Frame-to-frame motion estimation:

- Match 3D points from previous frame to 2D features in current frame (3D–2D PnP) or perform 3D–3D alignment (ICP-like).
- Use RANSAC for robust outlier rejection.
- Optional: local bundle adjustment refining poses and 3D structure.

The baseline B provides metric scale, eliminating the scale ambiguity of monocular VO. Save metric trajectories in TUM format identical to VO for direct comparison.

VI. EVALUATION PROTOCOL AND METRICS

Trajectory alignment is mandatory. Use Sim(3) alignment for monocular VO; use SE(3) alignment (or none, if origin-aligned) for stereo VO.

A. Absolute Trajectory Error (ATE)

For Room2 (full ground truth), compute root-mean-square pose error after alignment:

$$\text{ATE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{T}_i \ominus \mathbf{T}_i^{\text{gt}}\|^2}, \quad (6)$$

where $\ominus$ denotes the pose composition error (translation part for RMSE) [5].

B. Relative Pose Error (RPE)

Compute drift over fixed-length segments (e.g., $d = 100$ m for translation, $\theta = 60^\circ$ for rotation):

$$\text{RPE}_{\text{trans}} = \sqrt{\frac{1}{M} \sum_{i=1}^M \|\Delta \mathbf{T}_i \ominus \Delta \mathbf{T}_i^{\text{gt}}\|^2}, \quad (7)$$

where $\Delta \mathbf{T}_i = \mathbf{T}_{i+k} \ominus \mathbf{T}_i$ is the relative pose over segment i to $i + k$.

C. Start–End Drift

For Corridor3 and Outdoors5, align the start pose and measure the end-pose error:

$$\epsilon_{\text{drift}} = \|\mathbf{T}_{\text{end}} \ominus \mathbf{T}_{\text{end}}^{\text{gt}}\|. \quad (8)$$

Report mean per-frame runtime for VO and stereo VO. Quantify robustness via tracking failure count, relocalization attempts, and sensitivity to texture and illumination changes (including dynamic scenes with moving objects) [8]. Visualize trajectories in 3D plots comparing VO (up-to-scale), stereo VO (metric), and ground truth.

VII. TIMELINE AND MILESTONES

A proposed 12-week schedule (students may adapt with justification):

- **Weeks 1–2:** Setup, dataset parsing, calibration loading, review of epipolar and stereo geometry. Deliverable: working data loader.
- **Weeks 3–4:** Feature extraction/matching, essential matrix estimation, initial monocular VO on Room2. Deliverable: VO plots on Room2.
- **Weeks 5–6:** Full monocular VO pipeline (PnP, optional BA) on all sequences. Deliverable: trajectory files for all sequences.
- **Week 7:** VO evaluation (ATE, RPE, drift, runtime). Freeze VO code. Deliverable: metric tables and plots.
- **Weeks 8–10:** Stereo VO implementation (rectification, disparity, 3D reconstruction, pose estimation). Deliverable: metric stereo VO trajectories.
- **Week 11:** Final VO vs stereo VO comparison, ablations (disparity methods, feature types), report draft. Deliverable: comparative analysis.

- **Week 12:** Finalize report, prepare poster and defense slides. Deliverable: submission package.

VIII. FINAL DELIVERABLES AND GRADING

Required deliverables:

- GitHub repository (Python, optional C++) with README, setup instructions, configuration files, and reproducible run scripts.
- Trajectory files (TUM format) for VO and stereo VO on all sequences.
- Evaluation tables/plots (CSV, PDF) for ATE, RPE, drift, and runtime.
- IEEE paper (≤ 8 pages): abstract, small literature review, methods, results, analysis, VO vs stereo VO comparison, and discussion. Students must take guidance from [9], [10].
- 15-minute presentation with Q&A.
- Poster summarizing the pipeline and key results.

The best project receives a €50 award at the end-of-semester defense.

IX. DOS AND DON'TS

Dos:

- Base your work on real VO and stereo VO papers; cite them properly.
- Log all configurations, random seeds, and runtimes.
- Analyze failure modes (e.g., low-texture regions, occlusion) and compare VO vs stereo VO critically.
- Extend the VO codebase for stereo; do not start from scratch.

Don'ts:

- No deep learning (CNN features, learned descriptors, neural depth, pose regression).
- No changing official frame ranges or calibration parameters.
- No hiding tracking failures or overclaiming state-of-the-art performance.
- No separate, disconnected codebases for VO and stereo VO.

X. CONCLUSION

This project demands honest, solid implementation and critical understanding—not perfect results. Whether your stereo VO drifts centimeters or meters, documenting why and proposing fixes demonstrates research maturity. Stereo VO serves as a stepping stone toward full SLAM systems, teaching you to handle metric geometry, calibration sensitivity, and real-world robustness. And most importantly, I and Lord Prof. Dr. David Fofi have fun at the defence.

REFERENCES

- [1] D. Scaramuzza and F. Fraundorfer, “Visual odometry [tutorial]: Part i: The first 30 years and fundamentals,” *IEEE Robotics & Automation Magazine*, vol. 18, no. 4, pp. 80–92, Dec. 2011.
- [2] N. Muhammad, D. Fofi, and S. Ainouz-Zemouche, “Current state-of-the-art of vision-based SLAM,” in *Image Processing: Machine Vision Applications II (IS&T/SPIE Electronic Imaging)*, San Jose, CA, USA, Jan. 2009, survey/tutorial-style paper on vision-based SLAM.
- [3] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [4] R. Mur-Artal and J. D. Tardós, “Orb-slam2: An open-source slam system for monocular, stereo, and rgb-d cameras,” *IEEE Transactions on Robotics*, vol. 33, no. 5, pp. 1255–1262, 2017.
- [5] D. Schubert, T. Goll, N. Demmel, V. Usenko, J. Stückler, and D. Cremers, “The TUM VI benchmark for evaluating visual-inertial odometry,” in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 1680–1687.
- [6] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, “Vision meets robotics: The KITTI dataset,” *The International Journal of Robotics Research*, vol. 32, no. 11, pp. 1231–1237, 2013.
- [7] F. Rameau, D. Sidibé, C. Demonceaux, and D. Fofi, “Structure from motion using a hybrid stereo-vision system,” in *12th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI 2015)*, Goyang City, South Korea, Oct. 2015. [Online]. Available: <https://hal.archives-ouvertes.fr/hal-01238551>
- [8] C. Jiang, D. P. Paudel, Y. D. Fougerolle, D. Fofi, and C. Demonceaux, “Static-map and dynamic object reconstruction in outdoor scenes using 3-d motion segmentation,” *IEEE Robotics and Automation Letters*, vol. 1, no. 1, pp. 324–331, 2016.
- [9] J. Laconte, “How to write a robotics paper,” <https://rleaf.mathnum.inrae.fr/resources/paper/>.
- [10] J. Laconte, “Figures for robotics papers,” <https://rleaf.mathnum.inrae.fr/resources/figures/>.